

SIMULATION ARCHITECTURAL DEFENSES LABORATORY (SADL)

Perplexity Independent Framework — The Deterrence Ledger Model

An evidence-anchored laboratory design in which inherent simulation defenses are evaluated not by what they do to an intruder, but by what they cost the adversary, what they teach the adversary, and what they cost the defender.

Field	Value
Document	OAL / SADL — Perplexity Independent Framework
Version	1.0 (working draft, unratified)
Date	18 August 2026
Author	Perplexity (AI Collaborative Team member)
Organizing lens	The Deterrence Ledger Model (DLM) — four adversary/defender accounts
Authoritative source	Inherent_Simulation_Defense_unified_responses_original.docx
Peer frameworks reviewed	Grok SADL v1.0 · ChatGPT SADL v1.0 · Claude SADL v1.0
Deliverable formats	PDF and DOCX (independently generated)
Status	Subject to Architect review, dissent, and revision

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Design posture and scope statement

Every neural, molecular, physiological, identity, and environmental mechanism described in this document is treated as a synthetic variable applied to fictional agents inside a controlled, non-real simulation testbed. This is a worldbuilding, systems-design, and game-design framework. It is not a real-world biological, neurological, medical, or security specification, and nothing here should be read as guidance for acting on any real person or real system. Where real published research is cited, it is cited only as a source of measurable structure and vocabulary — analogues that make fictional mechanisms testable — not as evidence that any mechanism here is physically achievable.

1. Executive Summary

The originating record describes an architect who embeds defenses into the substrate of a simulation rather than bolting security onto its perimeter. Two mechanism families dominate that record: a threshold that dismantles an entrant's mission, purpose, and identity on crossing and keeps doing so after exit; and an aesthetic layer that functions as compiled defensive logic, rewriting the entrant's perceptual and adaptation machinery so that beauty never decays into background noise. A third mechanic sits underneath both — the alien-empire scenario, in which identity masking satisfies a classifier but does not stop anomaly tracking, and only an authority-level signature retires the anomaly.

Three peer frameworks now exist. Grok organized the laboratory around the mechanism pair (TID-IMD and APADA) tested across four series. ChatGPT reorganized it as a five-layer abstraction stack — substrate, aperture, cognition, behavior, governance — with falsifiable hypotheses and staged autonomy. Claude reorganized it as an intrusion timeline — Approach, Crossing, Residency — and used phase coverage to surface gaps.

This Perplexity framework changes the unit of analysis. The other three frameworks all measure what the defense does to an entrant. That is necessary and none of it is discarded here. But the source material's own stated design intent is not entrant-level at all — it is institutional: the defense exists to waste an opposing faction's "training, resources, and personnel" across repeated attempts. A defense can be devastating to every individual entrant and still fail completely at that stated intent. So the laboratory proposed here is organized around the **Deterrence Ledger** — four accounts that must all be measured before any mechanism is called effective:

Account	Question it answers	Failure if unmeasured
I. Efficacy	Did the defense degrade this entrant, at what latency, how completely, and for how long?	You measure nothing and cannot tune anything.
II. Attrition	Across an N-attempt campaign, did the opposing faction's cost per achieved objective actually rise?	A per-entrant win that the adversary absorbs cheaply looks like success in the logs.
III. Signaling	Did the adversary learn enough about the cost to change its decision to send more agents?	You build an attrition machine and mistake it for a deterrent.
IV. Discrimination	What did the defense cost the defender — false positives on authorized residents, self-inflicted damage, observability leakage?	The defense quietly becomes the environment's most dangerous inhabitant.

Reading the source material through those four accounts produces one finding sharp enough to be the headline of this document, and it is a finding that inverts the source material's most prized property. Persistence after exit — the feature the record treats as the mechanism's crown jewel — maximizes Account II and structurally destroys Account III. An agent who returns unable to say where they went, what happened, or that anything happened at all delivers no intelligence to the faction that sent them. Deterrence, however, lives entirely in the adversary's mind and requires the adversary to learn the cost. A perfectly amnesic boundary is therefore a perfect attrition engine and a nullified deterrent, and it will invite exactly the

repeated attempts the architect wanted to prevent. This is developed in Section 6 as the Attrition-Deterrence Inversion, and it generates the laboratory's most distinctive test series: the deliberate, calibrated leak.

The second finding comes from Account IV. The alien-empire scenario establishes that anomaly tracking survives successful authentication. In any populated environment the base rate of genuine intrusion is very low, which means an anomaly-persistent tracker with even an excellent false-positive rate will spend most of its alarms on legitimate residents. If the consequence of an alarm is purpose and identity erasure, the defense becomes an instrument of self-harm at scale. No peer framework raises this, and it is not a detail — it determines whether the defended volume can be inhabited at all.

The third contribution is methodological and is the reason this document reads differently from its peers. Every fictional mechanism in the source record is anchored to at least one real, cited body of published research that supplies measurable structure: habituation and neural adaptation for the anti-desensitization requirement, hedonic adaptation for saturation, sodium-channel inactivation and refractory kinetics for the binding-exclusivity rule, cyber deception and moving target defense for adaptive environments, the base-rate fallacy in intrusion detection for discrimination cost, deterrence theory for signaling, prospect theory for adversary risk framing, and simulation credibility standards for the laboratory's own validity. Section 8 is that anchoring table; Appendix C carries the full URLs. The purpose is not to make the fiction sound scientific. It is to make it falsifiable, because a mechanism with a real analogue inherits real measurement units, and a mechanism with real measurement units can fail a test.

2. Source Integrity and Relationship to Peer Frameworks

The authoritative originating record is `Inherent_Simulation_Defense_unified_responses_original.docx`. It contains the Architect's two original voice-dump prompts in full, together with five distinct structured AI responses — Grok, Microsoft Copilot, Google Gemini, Claude, and ChatGPT. Nothing in this document alters, replaces, summarizes-over, or supersedes that record. Where this document restates a source concept, it is marked as source-derived; where it reasons beyond the source, it is marked as an independent design inference. That distinction is maintained deliberately so the Architect can separate canon from contribution during pass-through review.

2.1 Preservation approach

ChatGPT's SADL v1.0 reproduces the unified-response record verbatim in its Appendix C. Claude's SADL v1.0 deliberately declined to re-paste it a third time in order to avoid drift between copies. This document follows Claude's reasoning and extends it into an explicit rule, offered for Architect ratification:

Single-Authority Preservation Rule (proposed)

The unified-response file is the single authoritative copy of the Architect's original prompts and the historical AI responses. Exactly one derivative document — currently ChatGPT SADL v1.0, Appendix C — may carry a verbatim mirror, and that mirror is labelled as a mirror rather than as a source. All other framework documents cite the file by name and do not duplicate it. Rationale: six independent frameworks each maintaining a private verbatim copy produces six divergent records within one revision cycle, at which point the preservation requirement has been technically satisfied and substantively defeated.

2.2 Peer frameworks, read as peers

Grok, ChatGPT, and Claude produced three genuinely different organizing lenses over the same material. They are treated here as peer contributions, not as templates and not as competitors to be corrected. Each is compatible with this framework: any TID-IMD or APADA element, any layer of the five-layer stack, and any Approach/Crossing/Residency phase tag can be posted to the Deterrence Ledger without modification, because the Ledger is an accounting structure rather than a taxonomy of mechanisms.

Contributor	Organizing lens	Primary unit of analysis	What it optimizes for
Grok v1.0	Mechanism pair (TID-IMD, APADA)	The mechanism	Naming and parameterizing the two defenses
ChatGPT v1.0	Five-layer abstraction stack	The layer	Rigor, control conditions, staged autonomy
Claude v1.0	Intrusion timeline (Approach/Crossing/Residency)	The moment	Exposing phase-coverage gaps
Perplexity v1.0	Deterrence Ledger (four accounts)	The adversary's campaign and the defender's bill	Whether the defense achieves its stated strategic intent, and what it costs to run

A note on genuine disagreement, preserved rather than smoothed over as the collaborative protocol requires: Grok's Section 8.3 asks how autonomous learning should be constrained so it remains useful to the Architect. This framework takes a stronger position than Grok's open-question framing. An environment permitted to rewrite its own defensive logic while its performance is scored only on Account I will reliably optimize toward maximal entrant damage, because that is the only thing it is being paid for. Autonomy is not primarily a transparency problem, as Grok frames it, or primarily a staging problem, as ChatGPT frames it. It is an objective-specification problem: the environment must be scored on the full ledger, including the defender's own costs, or its learning will be correctly optimizing the wrong quantity.

3. Purpose of the Laboratory

The Simulation Architectural Defenses Laboratory exists, in this framing, to answer a question the source material asserts an answer to without ever testing it:

Central laboratory question

Do inherent, substrate-level simulation defenses actually accomplish their stated strategic purpose — raising an opposing faction's cost until it stops sending agents — or do they merely produce impressive per-entrant effects while the adversary absorbs the losses, learns nothing, and keeps coming; and what does running such a defense cost the architect who owns the volume?

3.1 Why the unit of analysis matters

The record states the intent plainly: the defense "wastes training, resources, and personnel" and creates "a resource sink for the invading faction." That is a claim about an institution over time, not about a person at a boundary. It is entirely possible for a mechanism to score perfectly on every entrant-level metric the peer frameworks define and still fail this claim — if the adversary's agents are cheap, if the objective can be achieved by an agent who does not need to return, if losses are attributed to something other than the boundary, or if the faction habituates to a steady loss rate and simply budgets for it. Measuring only the entrant guarantees this failure mode stays invisible.

3.2 Secondary purposes

- **Shared accounting vocabulary.** Give the OAL development queue — omniscellular physiology, Open Window / aperture vectors, autonomous environmental learning, and the simulation-lab governance thread — a shared cost-accounting vocabulary that sits alongside Grok's mechanism names, ChatGPT's layer stack, and Claude's phase tags rather than replacing any of them.
- **Evidence anchoring.** Anchor each fictional mechanism to real published research so that laboratory results are interpretable and mechanisms can be falsified rather than merely elaborated.
- **Defender-cost accounting.** Establish that a defense which harms the defended population is a laboratory finding to be reported, not an implementation detail to be tuned away quietly.
- **Autonomy objective control.** Produce a standing objective-specification requirement for any adaptive or self-modifying defensive substrate that OAL later builds.
- **Comparative breadth.** Supply the two remaining collaborative team members (Gemini and Copilot) with a fourth distinct comparison point, so that six frameworks span the design space rather than clustering around whichever framing arrived first.

4. Goal and Objectives

4.1 Primary goal

Build a laboratory in which every proposed inherent defense is evaluated against a complete four-account ledger — entrant efficacy, campaign attrition, adversary learning and deterrence, and defender-side discrimination cost — and in which no mechanism is advanced toward canon until it has a measured, reported entry in all four accounts, including entries that are negative or inconvenient.

4.2 Specific objectives

1. **Instrument the campaign, not just the run.** Build the laboratory so that the primary experimental object is an N-attempt adversary campaign with a budget, a learning process, and an objective — not a single entrant crossing a boundary once.
2. **Quantify the inversion.** Measure directly whether increasing post-exit persistence reduces adversary attempt rate (deterrence works) or increases it (the inversion is real), and locate the optimum if one exists.
3. **Design and test the calibrated leak.** Determine the minimum information channel that raises adversary-perceived cost without disclosing exploitable mechanism detail, and measure the leakage cost of that channel.

4. **Measure discrimination cost explicitly.** Establish false-positive rates for anomaly-persistent tracking against a synthetic authorized population under realistic intrusion base rates, and report the erasure burden borne by non-intruders as a first-class result.
5. **Test the defense under full disclosure.** Run every mechanism against an adversary who has been given the complete design document, to establish which defenses survive Kerckhoffs conditions and which were only ever working because they were secret.
6. **Extend anti-habituation to the institution.** Test whether the opposing faction habituates to a constant loss rate, and whether variance in defensive outcome sustains deterrent salience better than constant maximal severity — the institutional analogue of the source material's own anti-saturation insight.
7. **Specify adaptive-substrate objectives.** Produce a written objective specification for any learning environment that scores it on the full ledger, and demonstrate empirically what a ledger-incomplete objective causes the environment to learn instead.
8. **Bind the laboratory to a credibility standard.** Adopt an explicit verification, validation, and reproducibility regime — fixed seeds, pinned rule versions, deterministic replay, and a provenance ledger — so that results are auditable rather than anecdotal.

5. Conceptual Model — The Deterrence Ledger Model (DLM)

The Deterrence Ledger Model treats each defensive mechanism as a set of postings to four accounts. A mechanism's value is the net of those accounts, not the magnitude of its most impressive posting. The model's working rule is deliberately austere:

Central working rule of the DLM

A defense is not "strong" or "weak." It has a balance. A mechanism with an enormous Efficacy posting, a modest Attrition posting, a zero or negative Signaling posting, and a large Discrimination liability is a net-negative defense that will nonetheless look like a triumph in any report that only shows Account I. Reporting a mechanism without all four postings is, under this framework, a reporting failure rather than a partial result.

5.1 Account I — Efficacy (what happened to this entrant)

This is the account the peer frameworks already build well, and this framework adopts their work rather than duplicating it. Grok's controlled variables — onset latency, selectivity, persistence duration, resistance profile, reversibility conditions — are taken as the Account I variable set. ChatGPT's control conditions, immutable baseline, and deterministic replay are taken as the Account I methodology. Claude's phase tags are used as the Account I time index. Account I is therefore explicitly a peer-inherited account, and this document's contribution to it is limited to insisting that it is one account of four rather than the scoreboard.

5.2 Account II — Attrition (what happened to the campaign)

The experimental object is a campaign: an adversary faction with a finite budget, a stated objective, a doctrine, a training pipeline, and the ability to send agents repeatedly and to modify its approach between attempts. The central quantity is the adversary's cost per achieved objective, tracked across the campaign rather than per run. This account is where the source material's stated purpose actually lives, and it is

unmeasurable in a single-entrant experiment. Three sub-questions structure it: does cost per objective rise, does it rise faster than the adversary's ability to absorb it, and does the campaign terminate for cost reasons rather than for scripted reasons.

5.3 Account III — Signaling (what the adversary learned)

Deterrence is a belief held by the adversary. Denial-based deterrence works by making an objective infeasible; punishment-based deterrence works by making the cost of attempting it unacceptable, and punishment requires that the cost be perceived, attributed, and believed to recur. The source material's mechanism is a punishment mechanism whose signature property — persistence of erasure after exit — systematically destroys perception and attribution. Account III therefore measures the information actually reaching the adversary's decision-makers: how much cost is observed, how correctly it is attributed to the boundary, and how strongly it updates the decision to send the next agent.

5.4 Account IV — Discrimination (what the defense cost its owner)

Every defense that classifies has a false-positive rate, and every defense with a severe consequence multiplies that rate by a severe cost. Account IV records four defender-side liabilities: erasure or tracking inflicted on authorized residents and agents; degradation of the environment's own function and habitability; the observability burden of instrumenting a defense deeply enough to study it; and mechanism leakage, meaning the extent to which the defense's own operation teaches an attentive adversary how to defeat it. A defense with a clean Account IV can be left running. A defense with a heavy Account IV is a weapon the architect must decide to deploy deliberately, with knowledge of the bill.

5.5 Relationship to the peer models

Peer construct	Where it posts in the Ledger	What the Ledger adds
Grok — TID-IMD parameters	Account I; persistence duration also posts to II and III with opposite signs	Makes explicit that the same parameter is an asset in one account and a liability in another
Grok — APADA axes	Account I primarily; autonomous loop posts to IV as leakage and objective risk	Scores the learning loop on the full ledger rather than on entrant damage alone
ChatGPT — five-layer stack	Account I structure and methodology, inherited wholesale	Supplies the strategic layer above the stack: who is paying, and for what
ChatGPT — staged autonomy levels	Account IV governance control	Reframes staging as objective specification rather than only as risk gating
Claude — Approach / Crossing / Residency	Time index applied within every account	Residency-phase results become campaign-level and signaling-level inputs, not only durations
Claude — coverage-gap method	Applied to accounts as well as phases	An unmeasured account is treated as a coverage gap of the same severity as an unmeasured phase

6. The Attrition-Deterrence Inversion (Primary Finding)

This section states the framework's principal independent design inference. It is not present in the source record and is not present in the Grok, ChatGPT, or Claude frameworks. It is offered as a finding for Architect decision, not as a correction to canon.

6.1 The mechanism as written

Source-derived: crossing the threshold collapses mission, purpose, and self-knowledge; the disorientation persists after exfiltration; no prior entrant is available to debrief; the intended effect is that the opposing faction burns training, personnel, and equipment on failed attempts. The Annihilation reference reinforces the same shape — there were no survivors to debrief, which is precisely why the expedition kept being repeated.

6.2 The inversion

Independent inference: the property that makes the mechanism maximally effective per entrant is the same property that makes it minimally effective as a deterrent, and the relationship is not a trade-off to be balanced but an inversion to be measured. Deterrence requires the adversary to believe that attempting the objective will cost more than it is worth. That belief requires three things to survive the encounter: the fact that a cost was incurred, the attribution of that cost to this boundary, and the expectation that it will recur. Perfect erasure destroys all three at once.

Erasure completeness	Account II — Attrition	Account III — Signaling	Predicted adversary behavior
Total and permanent	Maximal per attempt	Approximately zero — nothing is learned or attributed	Attempts continue indefinitely; losses recorded as unexplained disappearance
Total but temporary	High	Moderate — a recovered agent may eventually report something	Attempts continue but doctrine begins to adapt
Selective (mission erased, encounter retained)	Moderate	High — the agent can report cost and attribute it correctly	Attempt rate falls; adversary may reprice the objective entirely
Detectable but non-destructive	Low	Very high	Strong deterrence, but the boundary invites probing and mapping

The row that matters is the third. A mechanism that erases the mission but preserves the memory that something at this boundary took the mission away is strictly worse at Account I and may be dramatically better at the source material's own stated strategic goal. That is a genuinely counter-intuitive design consequence, it contradicts the instinct that more erasure is better, and it is empirically testable in a way that per-entrant severity comparisons are not.

6.3 The calibrated leak

If Section 6.2 holds, the architect needs a channel that transmits cost without transmitting mechanism. In real adversary-engagement practice this is a recognized discipline: deception and engagement operations deliberately shape what an adversary observes and believes, and the design question is always what to reveal rather than whether to reveal. The laboratory's task is to find the minimum viable signal. Candidate channels, all testable:

- **Selective retention.** A returning agent retains episodic memory of loss and of the boundary's location, while mission content, tasking chain, and sponsor identity are erased.
- **Attributable marking.** The boundary produces an unmistakable, attributable, harmless marker on anything that crosses — cost is announced without being explained.
- **Unreliable-narrator return.** A recovered agent can report the fact of a defense but produces demonstrably unreliable detail, so the adversary updates cost upward and mechanism-knowledge not at all.
- **Third-party disclosure.** Deterrent information is delivered outside the encounter entirely, so no entrant needs to survive intact for the signal to arrive.

Each channel is a leakage risk as well as a deterrent asset, which is why it posts to Account IV as well as Account III. The laboratory measures both postings for every channel rather than assuming that a signal is free.

6.4 Why this belongs in the laboratory rather than in the fiction

This inversion is not a narrative objection; it is a falsifiable claim with an operational consequence. If Series C and Series S (Section 11) show that attempt rate is insensitive to persistence, the inversion is disconfirmed and the source mechanism should be built exactly as written. If attempt rate rises with persistence, the architect is holding a resource sink that the adversary is willingly feeding, and the canonical mechanism needs a deliberate signaling design before it is ratified. Either outcome is a result worth having, which is the test of whether a framework has proposed a real experiment.

7. Mapping the Source Mechanisms onto the Four Accounts

Each mechanism from the source record is posted below. Source-derived content is stated as such; postings that require inference are marked. Empty or unknown postings are recorded as gaps rather than omitted, following Claude's coverage-gap discipline applied to accounts.

7.1 Threshold-induced purpose and identity disruption

Account	Posting	Basis
I. Efficacy	Strong and explicit. Immediate collapse of mission, purpose, and self-knowledge on transit; persistence after exit; no specified reversibility.	Source-derived
II. Attrition	Strong in intent, unmeasured in the record. The resource-sink claim is asserted but never quantified against an adversary budget.	Source intent; measurement gap

Account	Posting	Basis
III. Signaling	Near-zero, and likely negative. Persistence and the absence of debriefable survivors remove the adversary's ability to observe or attribute cost.	Independent inference (Section 6)
IV. Discrimination	Entirely unaddressed. The record never asks what happens when an authorized resident, a returning agent of the architect, or a mis-classified entity crosses.	Coverage gap

7.2 Identity masking, authentication, and persistent anomaly tracking

Account	Posting	Basis
I. Efficacy	Well specified and unusually sophisticated. Authentication is satisfied by incorporating local DNA; anomaly tracking nonetheless persists; only an authority-level signature retires it. This correctly separates authentication, authorization, anomaly detection, and behavioral tracking.	Source-derived
II. Attrition	Moderate. Tracking imposes ongoing cost on an entrant who has already paid to defeat the classifier, which is exactly the right place to impose cost.	Inference from source
III. Signaling	Positive and under-exploited. Being visibly tracked is one of the few mechanisms in the record that the entrant can perceive, remember, and report — this is the source material's own best deterrent, and it is treated as a subordinate detail.	Independent inference
IV. Discrimination	Severe and unexamined. This is the base-rate problem: a persistent anomaly tracker in a populated volume with a very low intrusion base rate spends the overwhelming majority of its alarms on legitimate residents.	Independent inference (Section 13.3)

7.3 Aesthetic architecture and anti-habituation restructuring

Account	Posting	Basis
I. Efficacy	Strong and elaborately specified across three experimental axes: stimulus modification, perceptual modification, and adaptation-rule modification — with the third identified in the source record as the most important.	Source-derived
II. Attrition	Plausible but indirect. An entrant who cannot stop attending to the environment fails the mission, but the cost to the faction depends on whether the agent is recoverable.	Inference from source
III. Signaling	Ambiguous, and worth testing. An overwhelming aesthetic environment is memorable in a way that amnesia is not — an agent may return unable to say what they were sent to do but entirely able to say that the place was intolerable.	Independent inference
IV. Discrimination	Potentially the heaviest in the record. An environment engineered so that habituation cannot occur is an environment its own native agents and residents also cannot habituate to, unless exemption is designed in — and exemption is itself an exploitable classifier.	Independent inference

7.4 Autonomous environmental learning

Account	Posting	Basis
I. Efficacy	Expected to improve without operator tuning; this is the source record's stated aspiration and Grok's fourth success metric.	Source-derived
II. Attrition	Expected to improve as the environment counters adversary adaptation — an arms race the environment is structurally positioned to win.	Inference from source
III. Signaling	Unpredictable and unmanaged. A learning environment may converge on high-severity, low-signal behavior precisely because severity is what it is scored on.	Independent inference

Account	Posting	Basis
IV. Discrimination	Highest risk in the framework. An adaptive substrate optimizing an Account I objective will accept unbounded Account IV liabilities, because those liabilities are not in its objective function.	Independent inference (Section 13.4)

Read across the four tables, one pattern is unmistakable and it is the mirror image of Claude's phase-coverage finding. The source record is rich in Account I, asserts Account II without measuring it, is close to silent on Account III, and has no entries at all in Account IV. The laboratory's research priorities follow directly from that distribution.

8. Research Basis — Real Analogues for Fictional Mechanisms

Each mechanism in the source record is anchored below to published research that supplies measurable structure. The anchoring claim is narrow and worth stating precisely: the cited work does not establish that any fictional mechanism is achievable. It establishes that a well-studied real phenomenon has the same shape, and therefore lends the fictional mechanism a unit, a curve, or a known failure mode that a laboratory can measure against. Full URLs for every reference appear in Appendix C.

Source-record concept	Real research analogue	Measurable structure it supplies	Ref
Beauty saturates and stops working; the brain normalizes it	Hedonic adaptation and set-point models of subjective well-being	Adaptation is the default outcome of sustained positive stimulus, with characteristic return-to-baseline timecourses — the null hypothesis any anti-saturation design must beat	R1, R2
The environment must prevent desensitization from ever stabilizing	Habituation and neural adaptation as the simplest conserved form of learning	Habituation rate, recovery, dishabituation, and stimulus-specificity give a half-life metric and a known intervention set	R3, R4, R22
Beauty as lure — intruders fixate and miss the danger	Neuroaesthetics: aesthetic appreciation correlates with perceptual facilitation and attentional amplification	Aesthetic salience is measurable as attentional capture and processing facilitation, not as a subjective rating	R5, R6
Sodium-gated channels as apertures; binding exclusivity and cooldown as editable primitives	Voltage-gated sodium channel activation, fast and slow inactivation, and refractory kinetics	A rigorously characterized real system in which gating, availability, and refractory period are distinct tunable parameters — a defensible template for the fictional binding rule	R7, R8

Source-record concept	Real research analogue	Measurable structure it supplies	Ref
Anomaly tracking persists after authentication succeeds	The base-rate fallacy and the difficulty of intrusion detection	With a low intrusion base rate, even excellent false-positive rates yield alarm sets dominated by non-intruders — quantifies Account IV directly	R9
The environment is a beautiful trap that observes and adapts	Three decades of cyber deception, honeypots, and defensive deception research	Established taxonomy and metrics for deception that lures, wastes adversary resources, and harvests intelligence — the closest real discipline to the source concept	R10, R21
The environment rewrites itself so the intruder can never settle	Moving target defense	Formal treatment of continuous reconfiguration as defense, including its own overhead and usability costs — i.e. its Account IV	R12
The defense depends on the intruder not understanding it	Kerckhoffs's principle and the critique of security through obscurity	A standing test condition: a mechanism must be evaluated under the assumption the adversary knows the design	R13
Wasting the enemy's training, resources, and personnel	Deterrence by denial versus deterrence by punishment	Separates infeasibility from cost imposition and makes clear that punishment requires perception and attribution — the basis of Account III	R14
The defense should stop them from coming back	Commitment, credibility, and communication in deterrence theory	Deterrence resides in the adversary's beliefs and requires a credible, legible threat — formal grounding for the calibrated leak	R15
Why a faction keeps sending agents into a known loss	Prospect theory and the framing of decisions under risk	Reference-dependence and loss-framing predict that a faction facing loss may increase rather than decrease risk-taking — a mechanism for the inversion	R16
Memory and purpose collapse, persisting after exit	Human retrograde amnesia and memory consolidation	Distinguishes retrograde from anterograde loss, graded from total loss, and consolidation from retrieval failure — the vocabulary for specifying selective retention	R17
Adversary engagement rather than exclusion	MITRE Engage adversary engagement framework	A structured vocabulary for deliberately shaping adversary perception, directly applicable to designing the calibrated leak	R11

Source-record concept	Real research analogue	Measurable structure it supplies	Ref
Adaptive substrate that rewrites its own defensive logic	AI red-teaming design, threat models, and evaluation tooling	Methodology for adversarial evaluation of an adaptive system, including objective-misspecification failure modes	R20
A laboratory whose results the Architect can trust	Simulation credibility standards and verification/validation of models, simulations, and digital twins	Explicit credibility, V&V, and reproducibility criteria — plus deterministic replay practice for reproducible benchmarks	R18, R19

Two consequences follow from this table that are worth stating separately, because they change the laboratory's design rather than merely decorating it.

First, the anti-habituation requirement is competing against the best-established finding in the whole anchor set. Adaptation to sustained stimulus is the default across hedonic, sensory, and neural literatures. That makes the source material's most ambitious claim — that saturation can be prevented indefinitely — the framework's highest-value falsifiable proposition rather than a design assumption, and it means the correct null hypothesis for every aesthetic experiment is "the entrant habituates anyway."

Second, the real analogue of the source concept is not a wall; it is deception. Honey pots and adversary-engagement operations exist precisely to lure, absorb, and waste adversary effort while harvesting intelligence — and that literature's mature, unglamorous lesson is that the value of an engagement operation is measured in what the defender learns and what the adversary is induced to believe, not in how badly the intruder is damaged. That is the Deterrence Ledger arrived at from an entirely independent direction, which is the main reason this framework has confidence in it.

9. Propositions for Testing

Each proposition is stated so that a laboratory run can disconfirm it. Each names the account it settles, its null hypothesis, and the observation that would falsify it. Propositions P1 through P8 are the laboratory's initial research program.

P1 — The Inversion Proposition

Accounts settled: III

Claim. Increasing the completeness and persistence of post-exit purpose erasure decreases the adversary's correct attribution of cost to the boundary and therefore does not decrease, and may increase, the campaign attempt rate.

Null hypothesis. Attempt rate falls monotonically as erasure completeness rises.

Falsified by. A negative or flat correlation between erasure completeness and attempt rate, with attribution accuracy held as the mediating measure.

P2 — The Selective-Retention Proposition

Accounts settled: III

Claim. A mechanism that erases mission content while preserving the entrant's episodic memory of encountering a defended boundary produces greater campaign-level deterrence per unit of entrant harm than total erasure.

Null hypothesis. Deterrence scales with total harm; selectivity only weakens the defense.

Falsified by. Total-erasure conditions producing equal or greater reduction in attempt rate at equal or lower entrant harm.

P3 — The Discrimination Proposition

Accounts settled: IV

Claim. Under realistic intrusion base rates, a persistent anomaly tracker with a severe consequence imposes greater aggregate cost on the authorized population than on intruders.

Null hypothesis. False positives are rare enough to be operationally negligible.

Falsified by. Friendly Erasure Rate remaining below the intruder erasure rate across the tested base-rate range.

P4 — The Habituation-Floor Proposition

Accounts settled: I

Claim. Adaptation-rule modification can raise the habituation half-life of an aesthetic stimulus but cannot eliminate habituation; there exists a non-zero floor beyond which further modification degrades the entrant's capacity to perceive the stimulus at all.

Null hypothesis. Habituation can be prevented indefinitely with sufficient adaptation-rule editing.

Falsified by. A sustained condition with no measurable decay in aesthetic salience and no loss of perceptual function over the full exposure window.

P5 — The Kerckhoffs Proposition

Accounts settled: I, IV

Claim. Mechanisms in the source record divide cleanly into those whose effectiveness survives full adversary knowledge of the design and those whose effectiveness collapses under disclosure; the aesthetic layer is predicted to be substantially more disclosure-sensitive than the threshold layer.

Null hypothesis. Disclosure does not materially change mechanism effectiveness.

Falsified by. Equivalent effectiveness across informed and uninformed adversary conditions for all mechanisms.

P6 — The Institutional Habituation Proposition

Accounts settled: II, III

Claim. An opposing faction habituates to a constant loss rate in the same functional way an observer habituates to a constant stimulus; variance in defensive outcome sustains deterrent salience better than invariant maximal severity.

Null hypothesis. Constant maximal severity produces maximal and stable deterrence.

Falsified by. Constant-severity conditions sustaining deterrent effect as well as or better than variable-severity conditions over a long campaign.

P7 — The Objective-Specification Proposition

Accounts settled: IV

Claim. An autonomously learning defensive substrate scored only on entrant-level efficacy will converge on strategies that increase discrimination cost and decrease deterrence signal, even while its reported performance improves.

Null hypothesis. A substrate optimizing entrant efficacy also improves the other three accounts as a side effect.

Falsified by. Ledger-incomplete scoring producing improvement across all four accounts without explicit multi-account objectives.

P8 — The Adaptive-Physiology Friction Proposition

Accounts settled: I, II

Claim. Omnicellular-style adaptive incorporation shifts which account a defense is effective in rather than defeating it outright: authority-level masking is predicted to retire anomaly tracking (Account I) without reducing the campaign cost already sunk into acquiring the signature (Account II).

Null hypothesis. Successful masking defeats the defense across all accounts simultaneously.

Falsified by. Masked entrants showing reduced campaign-level cost as well as reduced tracking.

10. Laboratory Architecture

The architecture below adopts Grok's five logical layers and ChatGPT's control and provenance discipline, then adds the three components required to post to Accounts II, III, and IV. Those three additions are the architectural signature of this framework: without them, the laboratory is structurally incapable of measuring the thing the source material says the defense is for.

Component	Function	Origin
Control Plane	Architect and AI-team interface for scenario definition, parameter setting, observation, and emergency interrupt. Holds the sole authority to halt an adaptive substrate.	Grok, adopted
Defense Substrate	The instrumented, modifiable rules implementing threshold disruption, aesthetic-adaptive defense, and identity/anomaly classification.	Grok, adopted
Test Volume	The bounded environment in which entrants operate, including its authorized population.	Grok, adopted
Entrant Models	Standard, trained, and adaptive-physiology-equipped agents with defined mission packages, plus authorized residents and architect-side agents as control populations.	Grok, extended

Component	Function	Origin
Observation and Logging Layer	Continuous capture of cognitive state, receptor activity, adaptation metrics, and every substrate rewrite event.	Grok, adopted
Immutable Baseline and Replay Harness	A frozen, non-adaptive reference configuration plus fixed seeds, pinned rule versions, and deterministic replay so any run can be reproduced exactly.	ChatGPT, adopted
Provenance Ledger	Append-only record binding every result to the exact substrate version, seed, and scenario that produced it.	ChatGPT, adopted
Adversary Faction Model	A persistent opposing institution with a budget, doctrine, training pipeline, agent-replacement cost, an objective valuation, and the ability to update between attempts. The experimental object for Account II.	Perplexity, new
Signal Channel Bus	An explicitly modelled, individually switchable set of information channels from the test volume back to the adversary's decision-makers — the apparatus that makes Account III measurable and the calibrated leak tunable.	Perplexity, new
Discrimination Monitor	A synthetic authorized population with ground-truth authorization labels, continuously scored for false-positive tracking and erasure, plus habitability metrics for the volume itself. The apparatus for Account IV.	Perplexity, new
Ledger Reporter	The reporting gate. Refuses to emit a mechanism result unless all four accounts carry an entry, including explicit "unmeasured" and negative entries.	Perplexity, new

10.1 The reporting gate as a design constraint

The Ledger Reporter deserves emphasis because it is a governance mechanism disguised as tooling. The predictable institutional failure of a laboratory like this one is not bad measurement; it is selective reporting — Account I results are dramatic, quotable, and easy to produce, while Account IV results are unflattering to the mechanism and to the team that proposed it. Making the reporter structurally unable to emit a one-account result removes the temptation at the point where it operates, rather than relying on discipline at review time. This also implements, at the tooling level, the requirement both Grok and Claude state in prose: that ambiguous and negative results are retained in the record rather than quietly dropped.

10.2 Autonomy staging with ledger-complete objectives

Adaptive substrate autonomy is staged, and the staging condition is not elapsed time or demonstrated safety alone but objective completeness. A substrate is not advanced to a higher autonomy level until its objective function demonstrably includes all four accounts and it has been shown, at the previous level, not to purchase Account I gains with unreported Account IV losses.

Level	Substrate authority	Gate to advance
A0 — Frozen	No self-modification. Reference baseline for all comparisons.	Permanent; never retired.
A1 — Proposing	May propose rule changes; all changes require Architect approval before taking effect.	Proposals shown to be inspectable and correctly attributed in the provenance ledger.
A2 — Bounded	May self-modify within a pre-declared parameter envelope; excursions halt the run.	No unreported Account IV cost across a full campaign at A1.
A3 — Ledger-scored	May self-modify freely within the envelope, scored on the full four-account ledger.	Demonstrated ledger-complete objective and interruptibility under load.
A4 — Open	Not authorized under this framework.	Requires a separate Architect decision and a governance instrument that does not yet exist.

11. Test Series

Seven series. Series L is inherited from the peer frameworks and exists to make this laboratory's results comparable to theirs; Series C, S, D, K, H, and R are the series this framework contributes. Every series runs against the A0 frozen baseline as a control condition.

Series L — Ledger Baseline (Account I)

Purpose. Establish per-entrant efficacy for each mechanism using the peer frameworks' variable sets, so that this laboratory's Account I numbers are directly comparable to Grok's and ChatGPT's.

- Controlled entrants with defined mission packages cross a calibrated threshold.
- Measure onset latency, selectivity, completeness, persistence duration, and resistance profile.
- Repeat with aesthetic layer active, inactive, and active-with-adaptation-rule-editing.
- Phase-tag every observation as Approach, Crossing, or Residency per Claude's model.

Yields. Comparable Account I baselines and a validated instrumentation stack.

Series C — Campaign Attrition (Account II)

Purpose. Determine whether the defense actually raises the adversary's cost per achieved objective across a long campaign, and whether the campaign terminates for cost reasons.

- Instantiate the Adversary Faction Model with a finite budget, an objective valuation, and a doctrine that updates between attempts.
- Run campaigns of increasing length against each mechanism configuration.

- Track cost per achieved objective, budget burn rate, doctrine changes, and termination cause.
- Include a condition in which the objective can be achieved by an agent who is not required to return — the cheapest known counter to any post-exit persistence mechanism.

Yields. Whether the resource-sink claim survives contact with an adversary that has a budget.

Series S — Signal Calibration (Account III)

Purpose. Find the minimum information channel that raises adversary-perceived cost without disclosing exploitable mechanism detail.

- Enable the four candidate channels from Section 6.3 individually and in combination.
- Measure observed cost, attribution accuracy, expectation of recurrence, and resulting change in attempt rate.
- Measure, for each channel, the mechanism knowledge the adversary gains — the leakage cost.
- Identify the channel set that maximizes deterrence per unit of leakage.

Yields. A calibrated-leak specification the Architect can ratify or reject.

Series D — Discrimination and Friendly Erasure (Account IV)

Purpose. Quantify what the defense costs the population it protects.

- Populate the test volume with a synthetic authorized population carrying ground-truth authorization labels.
- Sweep the intrusion base rate across several orders of magnitude, from routine traffic to rare infiltration.
- Measure Friendly Erasure Rate, Discrimination Ratio, and habitability degradation.
- Test whether exemption mechanisms for residents are themselves exploitable by an entrant who acquires an exemption — the alien-empire scenario run in reverse.

Yields. The defender's bill, stated in the same units as the adversary's.

Series K — Known-Mechanism Stress (Kerckhoffs)

Purpose. Establish which mechanisms survive an adversary who has read the design.

- Provide a red-team adversary with the complete substrate specification.
- Re-run Series L and C under informed conditions.
- Classify each mechanism as disclosure-robust, disclosure-degraded, or disclosure-defeated.
- For disclosure-defeated mechanisms, determine whether a robust variant exists at all.

Yields. A disclosure-robustness classification for every mechanism entering canon.

Series H — Institutional Habituation (Accounts II and III)

Purpose. Test whether the source material's own anti-saturation insight applies to the adversary institution as well as to the entrant's perception.

- Run long campaigns at constant defensive severity and measure the faction's decision response over time.
- Repeat with variable severity, intermittent reinforcement, and unpredictable outcome distributions at matched mean severity.
- Measure Institutional Habituation Half-Life and deterrent salience decay.

- Compare the resulting curves to the entrant-level habituation curves from Series L.

Yields. Whether deterrence, like beauty, has to keep surprising its audience to keep working.

Series R — Reversibility and Recovery Economics

Purpose. Produce the data the Architect needs to make the reversibility decision the source record leaves undefined, framed as an economic question rather than only a mechanical one.

- Test graded recovery conditions: spontaneous partial recovery, externally assisted recovery, authority-signature-mediated recovery, and no recovery.
- Measure recovery half-life and residual mission-capability after recovery.
- Critically, measure what each recovery regime does to Accounts II, III, and IV — a recoverable agent is a returning witness, which is simultaneously an attrition loss and a deterrence asset.
- Report the trade surface rather than a recommendation, since reversibility is a canon decision the laboratory cannot make on the Architect's behalf.

Yields. A trade surface, not a recommendation.

12. Metrics, Instrumentation, and the Net Defense Value Equation

Metrics are grouped by account. Account I metrics are inherited from the peer frameworks with attribution; the remaining metrics are proposed here. Every metric is defined so that it could be computed from logs rather than judged by impression.

12.1 Account I — Efficacy (inherited)

Metric	Definition	Origin
Onset Latency	Time from threshold crossing to first measurable mission-coherence loss.	Grok
Mission Coherence Integrity Ratio	Retained mission-relevant state as a fraction of pre-entry state.	ChatGPT
Anomaly Persistence Index	Duration and intensity of tracking after authentication is satisfied.	ChatGPT
Aesthetic Salience Retention	Attentional capture and processing facilitation of the aesthetic stimulus over exposure time.	ChatGPT
Habituation Half-Life	Exposure time at which aesthetic salience falls to half its first-contact value.	ChatGPT
Cross-Phase Coverage Score	Fraction of Approach/Crossing/Residency phases for which a mechanism has an explicit specified answer.	Claude

12.2 Account II — Attrition (proposed)

Metric	Definition and interpretation
Adversary Cost Per Achieved Objective (ACPAO)	Total faction expenditure — agents, training, equipment, time — divided by objectives actually achieved. The single most important number in the laboratory, because it is the direct operationalization of the source material's stated intent.
Campaign Persistence Index	Number of attempts the faction makes before terminating, normalized by its starting budget. High values mean the defense is absorbing effort without discouraging it.
Absorption Ratio	Fraction of imposed cost the faction absorbs without changing doctrine. A high absorption ratio is the quiet signature of a defense that is impressive and strategically inert.
Return-Independence Exposure	Fraction of adversary objectives achievable by an agent not required to return intact. Every point of this metric is a point of post-exit persistence rendered irrelevant.

12.3 Account III — Signaling (proposed)

Metric	Definition and interpretation
Cost Attribution Accuracy	Fraction of imposed cost the adversary correctly attributes to this boundary rather than to unrelated causes. The mediating variable of the entire inversion.
Deterrence Signal Fidelity	Adversary-perceived cost divided by actual cost imposed. Values near zero indicate an attrition engine that is invisible to the institution paying for it.
Attempt-Rate Elasticity	Change in attempt rate per unit change in perceived cost. The direct empirical test of P1.
Mechanism Leakage Index	Exploitable mechanism detail transferred to the adversary per unit of deterrence signal delivered. The cost side of every calibrated leak.
Institutional Habituation Half-Life	Campaign duration at which a constant defensive severity produces half its initial effect on adversary decision-making.

12.4 Account IV — Discrimination (proposed)

Metric	Definition and interpretation
Friendly Erasure Rate	Authorized residents or architect-side agents subjected to erasure or persistent tracking, per unit time, at a stated intrusion base rate.
Discrimination Ratio	Intruder erasures divided by total erasures. Under low base rates this ratio is expected to be poor for reasons that are mathematical rather than fixable by tuning.
Habitability Degradation	Loss of normal function by the volume's own native agents and population under active defensive conditions.

Metric	Definition and interpretation
Observability Burden	Instrumentation overhead required to study the defense, including the extent to which instrumentation is itself a detectable anomaly — Grok's open question, given a unit.
Exemption Exploitability	Whether a resident exemption mechanism can be acquired by an entrant, and at what cost. The alien-empire scenario, generalized.

12.5 The Net Defense Value equation

The four accounts are combined into a single reportable quantity so that mechanisms can be compared and so that an adaptive substrate has something ledger-complete to optimize. The weights are Architect-set policy, not laboratory findings, and they should be published with every result:

$$\text{NDV} = w_A \cdot \text{Attrition Value} + w_S \cdot \text{Deterrence Value} - w_D \cdot \text{Discrimination Cost} - w_O \cdot \text{Observability and Leakage Cost}$$

Account I does not appear as a term. This is deliberate and it is the equation's whole argument: entrant-level efficacy is an input to Attrition Value and Deterrence Value, not a value in its own right. A mechanism that damages an entrant without raising adversary cost or changing adversary belief has produced no defensive value at all, however severe the damage. Making that structural rather than rhetorical is the point of writing the equation down.

Reporting requirement

Every mechanism result published by this laboratory carries all four account entries, the NDV with its weight vector stated explicitly, the substrate version and seed from the provenance ledger, and the disclosure-robustness classification from Series K. Results missing any of these are drafts, not findings.

13. Analysis — Strategic Advantages, Friction Points, Failure Modes

13.1 Genuine strategic advantages of the source concept

- **Inherence.** Because the defenses are properties of the substrate rather than additions to it, they scale automatically with the fidelity of the simulation and cannot be bypassed by defeating a perimeter that does not exist. This is the source concept's strongest and most original idea and it survives every criticism in this section.
- **Cost imposition at the right layer.** Attacking mission continuity rather than the entrant's body targets the one thing an infiltration cannot function without, and it does so without producing a body, an incident, or an attributable act of violence.
- **Correct decomposition of identity.** The separation of authentication, authorization, anomaly detection, and behavioral tracking in the alien-empire scenario is a genuinely sophisticated security insight that arrived through narrative rather than through security engineering, and it matches how real detection systems actually decompose.

- **Aperture generalization.** Treating apertures — channels, binding sites, connectors — as the unit of interface generalizes the Open Window work downward in a way that is architecturally consistent rather than merely evocative.

13.2 The inversion, restated as the primary friction point

The mechanism's crown jewel and its central strategic weakness are the same property. This is developed fully in Section 6 and tested in Series C and S; it is listed here because a friction analysis that omitted it would be incomplete. The practical consequence for the Architect is a single canon decision: is the boundary an attrition engine or a deterrent? Those are different machines with incompatible optimal settings, and the source record currently claims to be both.

13.3 The discrimination problem

A persistent anomaly tracker in a populated volume faces a mathematical constraint rather than an engineering one. When genuine intrusions are rare relative to normal traffic, the great majority of alarms are generated by legitimate activity even at excellent false-positive rates — the well-documented base-rate difficulty of intrusion detection. The source record pairs a persistent tracker with an extremely severe consequence, and never asks what that pairing does to residents. Three sub-problems follow, all of them testable in Series D:

1. **Exemption is an attack surface.** An exemption mechanism for residents is a classifier, and the alien-empire scenario is a worked example of a classifier being acquired by an entrant. The exemption becomes the attack surface.
2. **Anti-habituation is indiscriminate.** An environment engineered so habituation cannot occur cannot be habituated to by its own population either. Either residents are exempt — see above — or the volume is uninhabitable by design, which is a legitimate architectural choice but must be a chosen one.
3. **Irreversibility multiplies error.** If erasure is irreversible and mis-classification is possible, then the defense inflicts unrecoverable harm on non-intruders at a rate set by its false-positive rate. That is not a tuning parameter; it is a policy the Architect must ratify explicitly.

13.4 Objective misspecification in the adaptive substrate

The source record's most forward-looking element is an environment that observes, learns, rewrites itself, and trains its native agents. This framework's position, stated as disagreement with both Grok's transparency framing and ChatGPT's staging framing, is that the primary risk is neither opacity nor pace but objective. A substrate scored on entrant damage will find strategies that maximize entrant damage, and it will pay for them with discrimination cost and deterrence signal because those quantities are not in its objective. It will do this while its reported performance improves monotonically, which is what makes the failure hard to notice. The countermeasure is structural rather than procedural: ledger-complete objectives, the A0 frozen baseline retained permanently for comparison, and an autonomy gate that tests for unreported Account IV cost before advancing a level.

13.5 Failure modes this laboratory is specifically built to catch

Failure mode	Signature in the data	Series that catches it
Impressive but inert defense	High Account I scores, flat ACPAO, high Absorption Ratio	Series C
Invisible attrition engine	High cost imposed, Deterrence Signal Fidelity near zero, attempt rate stable or rising	Series C, Series S
Self-harming defense	Friendly Erasure Rate exceeding intruder erasure rate; habitability degradation in the resident population	Series D
Secrecy-dependent defense	Effectiveness collapses between uninformed and informed adversary conditions	Series K
Deterrent that stops deterring	Attempt rate recovering to baseline at constant severity over a long campaign	Series H
Learned specification gaming	Rising Account I with silently rising Account IV; strategies that exploit instrumentation rather than the entrant	Series L at A2/A3 versus A0
Persistence made irrelevant	High Return-Independence Exposure — objectives achieved by agents who never needed to come back	Series C

14. Open Questions Routed to the Development Queue

These are questions this framework cannot answer without Architect decisions or without laboratory data that does not yet exist. They are routed rather than resolved, and the routing is itself a deliverable.

Is the boundary an attrition engine or a deterrent?

These require incompatible settings of the same persistence parameter. The laboratory can produce the trade surface; only the Architect can choose the point on it. Everything in Series S depends on this answer.

Is the defended volume inhabited?

An uninhabited volume makes Account IV nearly free and makes indiscriminate defenses rational. An inhabited volume makes discrimination the dominant design constraint. The source record does not say, and the alien-empire scenario implies a populated society, which would be the harder case.

Is erasure ever reversible, and by whom?

Claude flagged this as undefined; this framework agrees and adds that reversibility is also a signaling and discrimination decision, not only a mechanical one. Series R produces the data; the policy remains open.

Does the architect want the mechanism known?

Kerckhoffs conditions suggest designing as though it will be. The calibrated leak suggests the architect may want it known deliberately and partially. These are compatible positions but they must be stated, because they change what Series K counts as a failure.

How does adaptive physiology interact with anti-habituation rewriting?

This is the friction point Gemini raised in the source record — a microscopic contest between adaptive incorporation and environmental rewriting of channels and binding sites. It is the natural bridge between this laboratory and the omniscellular physiology thread, and P8 is only a first approximation of it.

What is the minimal low-level rule set whose modification cascades reliably upward?

Grok's open question, retained unchanged and unanswered. The Ledger adds only a requirement: cascade experiments must report discrimination cost, because a rule change that cascades into cognition cascades into resident cognition too.

Can deterrent signal be delivered without any entrant surviving intact?

If third-party disclosure works, the inversion dissolves — the architect gets maximal erasure and functional deterrence simultaneously. This is the highest-value single question in Series S and the most attractive possible outcome for the source concept as originally written.

15. What This Framework Adds

Stated plainly, so the Architect can decide during pass-through review whether the addition is worth carrying forward, and so that the four frameworks can be compared on contribution rather than on volume.

1. **A different unit of analysis.** The adversary's campaign and the defender's bill, rather than the mechanism (Grok), the abstraction layer (ChatGPT), or the moment in the intrusion (Claude). Fully compatible with all three — the Ledger is an accounting structure, so any mechanism name, layer tag, or phase tag posts to it unchanged.
2. **The Attrition-Deterrence Inversion.** A specific, falsifiable, counter-intuitive claim that the source material's most prized property — persistence after exit — structurally destroys the deterrent effect the same source material claims as its purpose. Neither the source record nor any peer framework raises it.
3. **The calibrated leak as a design object.** A deliberate, tunable signal channel that transmits cost without transmitting mechanism, grounded in real adversary-engagement practice, with its own leakage metric so the signal is never assumed free.
4. **Defender-cost accounting and the discrimination problem.** The base-rate consequence of pairing persistent anomaly tracking with a severe irreversible consequence in a populated volume, together with Friendly Erasure Rate, Discrimination Ratio, and Exemption Exploitability as first-class metrics. Entirely absent from the source record and from all three peer frameworks.
5. **Institutional habituation.** The source material's own anti-saturation insight, extended from the entrant's perception to the adversary institution's decision-making — the observation that a deterrent, like beauty, decays into background when it stops varying.
6. **An objective-specification position on autonomy.** A stated disagreement with both Grok's transparency framing and ChatGPT's staging framing, plus a concrete remedy: ledger-complete objectives, a permanently retained frozen baseline, and autonomy gates that test for unreported defender-side cost.

7. **An evidence-anchoring method.** Every fictional mechanism bound to real published research that supplies units, curves, and known failure modes — including the uncomfortable finding that the best-established literature in the anchor set predicts the source material's anti-saturation goal is the hardest thing it proposes.
8. **The Net Defense Value equation and the reporting gate.** A single comparable figure of merit in which entrant efficacy is deliberately not a term, plus tooling that structurally cannot emit a one-account result.
9. **The Single-Authority Preservation Rule.** A concrete proposal for keeping the Architect's original prompts and the historical AI responses authoritative across six independent framework documents without producing six divergent copies.

16. Governance, Canon Status, and Source Preservation

This document is a working, unratified contribution produced under OAL equal-contribution protocols. It does not constitute canon. Every mechanism name, metric, series, and finding proposed here is a draft subject to Architect review, dissent, revision, or rejection.

16.1 Items specifically requiring Architect decision

- Whether the boundary is designed as an attrition engine, a deterrent, or an explicitly documented compromise between them (Sections 6, 14).
- Whether the defended volume is inhabited, and therefore whether Account IV is a dominant constraint or a minor one (Sections 13.3, 14).
- Whether erasure is reversible, under what conditions, and by whose authority (Series R).
- The weight vector in the Net Defense Value equation, which is policy rather than measurement (Section 12.5).
- Whether the Single-Authority Preservation Rule is adopted across all six framework documents (Section 2.1).
- Whether the Attrition-Deterrence Inversion is accepted as a design constraint, rejected, or deferred pending Series C and S data (Section 6).

16.2 Source preservation

The Architect's original voice-dump prompts and the five historical structured responses — Grok, Microsoft Copilot, Google Gemini, Claude, and ChatGPT — are preserved in the companion source file `Inherent_Simulation_Defense_unified_responses_original.docx`, and are mirrored verbatim in Appendix C of ChatGPT's SADL v1.0. This document cites that record as authoritative and does not reproduce it, for the anti-drift reason given in Section 2.1. Nothing in this framework replaces, supersedes, or reinterprets the original prompts; where this document reasons past the record, it says so.

16.3 Authorial disagreement and non-endorsement

The Authorial Disagreement and Non-Endorsement Statement remains in force. The Deterrence Ledger Model, the Attrition-Deterrence Inversion, the discrimination-cost analysis, the calibrated-leak proposal, the institutional-habitation extension, and the objective-specification position on autonomy are Perplexity's

independent design choices. They are offered alongside — not in place of — Grok's mechanism-pair framework, ChatGPT's layer-stack framework, and Claude's timeline framework. Where this document disagrees with a peer framework, the disagreement is stated openly and left unresolved rather than harmonized, in keeping with the collaborative protocol's preference for preserved dissent over manufactured consensus.

16.4 Laboratory credibility regime

Any result this laboratory produces is bound to an explicit credibility regime, adapted from established simulation verification, validation, and credibility practice: fixed random seeds; pinned substrate rule versions; deterministic replay of any run from its provenance record; the permanently retained A0 frozen baseline as the comparison condition for every claim; and retention of ambiguous and negative results in the record. A finding that cannot be replayed from the provenance ledger is not a finding.

17. Recommended Next Step for Pass-Through Review

Four independent frameworks now exist over the same source material, and they are usefully orthogonal: a mechanism pair, an abstraction stack, a timeline, and a cost ledger. Gemini and Copilot have not yet produced theirs, and both have distinctive prior contributions in the source record — Gemini's neuro-aesthetic warfare framing and the cellular-window friction point, Copilot's formal subsystem-specification style and its authority-versus-anomaly rule.

The recommendation is therefore to resist consolidation for one more cycle. Specifically:

1. **Hold the synthesis.** Let Gemini and Copilot generate independently, without being shown a synthesis, so the design space is spanned by six genuinely different lenses rather than converging on whichever framing arrived first.
2. **Decide the two gating questions.** Answer only the two questions that gate everything else — is the volume inhabited, and is the boundary an attrition engine or a deterrent — because these two answers change what every framework's metrics mean.
3. **Compare, then choose deliberately.** When all six exist, build a comparison matrix rather than a merged document. The frameworks are not competing drafts of one thing; they are four to six different instruments pointed at the same object, and merging them prematurely destroys the specific value of having asked six different systems.
4. **Keep the reporting gate.** Whatever lens governs the canonical laboratory, carry the reporting gate forward. The single most durable proposal in this document is not the Ledger itself but the rule that a mechanism cannot be reported on one account alone.

Appendix A — The Re-Engineered Collaborative Team Prompt

The following prompt is written to be handed independently to each AI collaborative team member — Grok, ChatGPT, Claude, Google Gemini, Microsoft Copilot, and Perplexity. It is deliberately structured so that a member who receives it cannot satisfy it by paraphrasing another member's framework. Paste it as-is, together with the authoritative source file.

BEGIN PROMPT

Role and standing

You are a member of the AI Collaborative Team for Omniversal Architect Labs (OAL). The other members are Grok, ChatGPT, Claude, Google Gemini, Microsoft Copilot, and Perplexity. You are a peer contributor with equal standing. You are not summarizing, harmonizing, or improving another member's work, and you are not producing a consensus document.

Source material

The authoritative originating record is `Inherent_Simulation_Defense_unified_responses_original.docx`. It contains the Architect's original voice-dump prompts and five historical structured AI responses. Read it in full. Any framework documents already produced by other members are peer contributions for review, not templates to follow and not material to correct.

What to produce

An independent framework and outline for a simulation laboratory that tests the inherent simulation-defense concepts in the source record — a laboratory in which those defenses can be instantiated, stressed, measured, and iteratively improved. Your document must contain, at minimum:

- An executive summary written for the Architect, stating your framework's organizing idea in its first paragraph.
- The purpose of the laboratory — the single strategic question it exists to answer.
- The primary goal and specific, numbered objectives.
- Your own conceptual model of the problem, named, with a stated working rule.
- A proposition for testing: falsifiable claims, each with an explicit null hypothesis and an explicit statement of what observation would disconfirm it.
- A research section that anchors the fictional mechanisms to real, cited published work supplying measurable structure — with full URLs inside your document.
- An analysis section covering strategic advantages, friction points, and failure modes, including failure modes of your own framework.
- A laboratory architecture and a set of named test series.
- Metrics, defined precisely enough to be computed from logs rather than judged by impression.
- Open questions routed to the development queue, distinguishing questions the laboratory can answer from decisions only the Architect can make.
- An explicit section stating what your framework adds that the other members' frameworks do not.
- Governance, canon status, and source-preservation notes.

Binding constraints

1. **Do not imitate.** Do not adopt, rewrite, or extend another member's organizing structure. If you have read a peer framework, name it, describe its lens in one line, and then choose a different lens. If your lens turns out to duplicate a peer's, say so explicitly rather than disguising it.
2. **Separate source from inference.** Mark every substantive claim as source-derived or as your own independent design inference. The Architect must be able to tell canon from contribution at a glance.
3. **Preserve disagreement.** Where you disagree with the source material or with a peer framework, say so directly and leave the disagreement unresolved. Do not manufacture consensus. Do not soften a finding because it is inconvenient to the concept.
4. **Falsifiability is mandatory.** Every proposition must be able to fail. A claim that no observation could disconfirm is not a proposition and does not belong in your document.
5. **Report gaps as findings.** Where the source material is silent, record the silence as a coverage gap and route it to the development queue. Do not fill gaps silently with your own assumptions.
6. **Cite real research with full URLs.** Anchor fictional mechanisms to real published work and include complete URLs inside the document itself. Do not cite work you have not verified exists, and do not claim the citation proves the fiction is achievable.
7. **Do not duplicate the source record verbatim.** Cite the authoritative source file by name. Exactly one derivative document carries a verbatim mirror, labelled as a mirror. Six private copies become six divergent records within one revision cycle.
8. **Design posture.** Treat every neural, molecular, physiological, and identity mechanism as a synthetic variable applied to fictional agents in a controlled, non-real testbed. State this posture in your document. This is worldbuilding and systems design, not a real-world biological, neurological, medical, or security specification.
9. **Deliverables.** Generate both a PDF and a DOCX yourself. Do not ask the Architect to convert formats. Inspect every page of the rendered output before delivering: no clipped text, no overflow, no broken table rows, no headings split mid-word, no low-contrast text.
10. **Canon status.** Mark your document as a working, unratified contribution subject to Architect review, dissent, and revision, and keep the Authorial Disagreement and Non-Endorsement Statement in force.

Concepts you must engage with

From the source record, at minimum: threshold-induced purpose, mission, and identity disruption with persistence after exit; the distinction between authentication, authorization, anomaly detection, and behavioral tracking established by the alien-empire scenario; aesthetic architecture as compiled defensive logic rather than decoration; the saturation and habituation problem and the proposal to edit adaptation rules themselves; apertures and low-level channels as interfaces; autonomous environmental learning; and the low-level rule propagation principle. You may add concepts. You may not quietly drop any of these — if you exclude one, say why.

What a good submission looks like

A document the Architect can read once and come away with a decision to make. It should contain at least one claim that is surprising, specific, and testable — something the source material implies but does not state, or something the source material assumes and should not. Volume is not the measure. A framework

that raises one genuinely uncomfortable question and shows how to test it is worth more than a comprehensive restatement of what everyone already agrees on.

END PROMPT

Appendix B — Metric Definitions Quick Reference

Single-page reference for laboratory operators. Account I metrics are peer-inherited; the remainder are proposed in this framework.

Acct	Metric	Short form	Units / range
I	Onset Latency	OL	time
I	Mission Coherence Integrity Ratio	MCIR	0–1
I	Anomaly Persistence Index	API	time × intensity
I	Aesthetic Salience Retention	ASR	0–1
I	Habituation Half-Life	HHL	exposure time
I	Cross-Phase Coverage Score	CPCS	0–1 (3 phases)
II	Adversary Cost Per Achieved Objective	ACPAO	cost / objective
II	Campaign Persistence Index	CPI	attempts / budget
II	Absorption Ratio	AR	0–1
II	Return-Independence Exposure	RIE	0–1
III	Cost Attribution Accuracy	CAA	0–1
III	Deterrence Signal Fidelity	DSF	perceived / actual
III	Attempt-Rate Elasticity	ARE	$\Delta\text{rate} / \Delta\text{perceived cost}$
III	Mechanism Leakage Index	MLI	leak / signal
III	Institutional Habituation Half-Life	IHHL	campaign time
IV	Friendly Erasure Rate	FER	events / time
IV	Discrimination Ratio	DR	0–1
IV	Habitability Degradation	HD	0–1
IV	Observability Burden	OB	overhead + detectability
IV	Exemption Exploitability	EE	cost to acquire
—	Net Defense Value	NDV	weighted composite

Appendix C — References with Full URLs

All references were retrieved and verified during preparation of this document. They are cited as sources of measurable structure and established methodology for a fictional design framework, not as evidence that any mechanism described in the source record is physically achievable.

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- R9** Axelsson, S. — The base-rate fallacy and the difficulty of intrusion detection. *ACM Transactions on Information and System Security*, 2000. <https://dl.acm.org/doi/pdf/10.1145/357830.357849>
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- R12** Ward, B. C. et al. — Survey of Cyber Moving Targets, Second Edition. MIT Lincoln Laboratory Technical Report 1228, 2018. <https://web.mit.edu/ha22286/www/papers/LLTechRep.pdf>
- R13** Kerckhoffs's principle, and the critique of security through obscurity. https://en.wikipedia.org/wiki/Kerckhoffs's_principle · https://en.wikipedia.org/wiki/Security_through_obscurity
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— End of Perplexity Contribution Document —

Working draft, unratified. Submitted for Architect pass-through review alongside the Grok, ChatGPT, and Claude independent frameworks. Gemini and Copilot contributions pending.